

MAINCRAFTS TECHNOLOGY

VLSI Design Internship: Task 3 Completion Report

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Platform Used	Icarus Verilog + GTKWave	Submission Date	August 17, 2026

1. EXECUTIVE OBJECTIVE

The core objective of Task 3 is to transition from combinational logic design into sequential logic elements using Register Transfer Level (RTL) modeling in Verilog HDL. This report documents the implementation, testbench creation, console simulation results, and visual verification for:

- D Flip-Flop
- JK Flip-Flop
- 4-bit Register
- 4-bit Parameterized Counter

2. RTL DESIGN CONCEPTS & VERILOG FUNDAMENTALS

- Sequential Logic:** Logic blocks where the output depends on both present inputs and past states, controlled by a clock signal utilizing the `always @(posedge clk)` construct.
- Synchronous Resets:** Reset conditions evaluated only at the active edge of the clock, preventing metastability and ensuring clean state transitions.
- Parameterization:** Utilizing parameters (e.g., `#(parameter WIDTH = 4)`) to create scalable and highly reusable hardware modules.

3. CIRCUIT IMPLEMENTATIONS, SIMULATION & RESULTS

Part A: D Flip-Flop

1. RTL Block Diagram

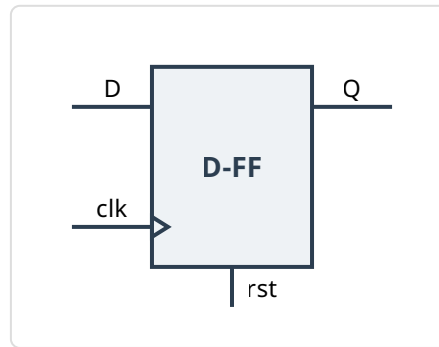


Fig 1: RTL Block Diagram representing the inputs and outputs of the D Flip-Flop.

2. Verilog RTL Design & Simulation Console (VS Code)

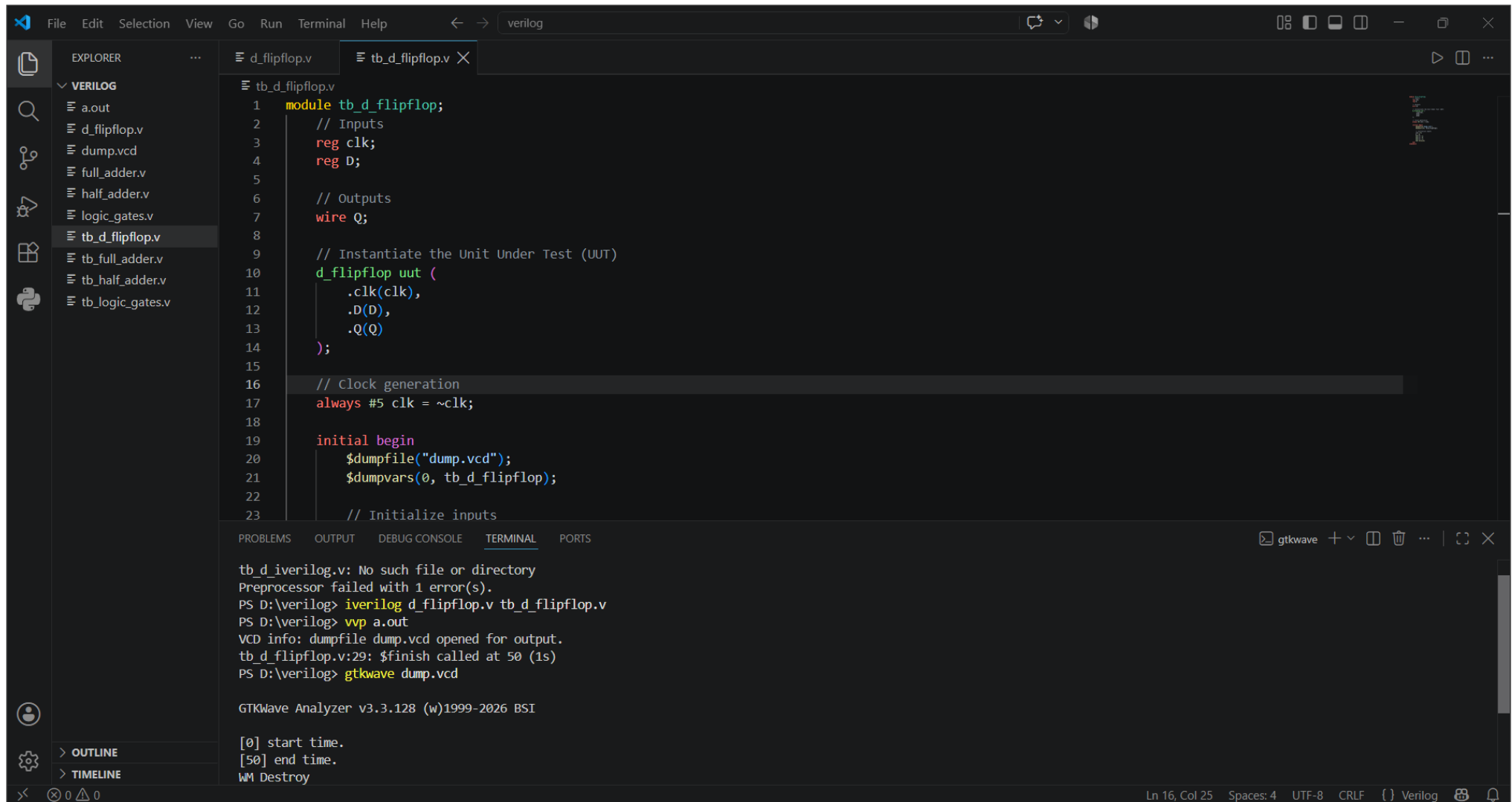


Fig 2: Module definition, Testbench instantiation, and Terminal output for D Flip-Flop.

3. GTKWave Waveform Verification

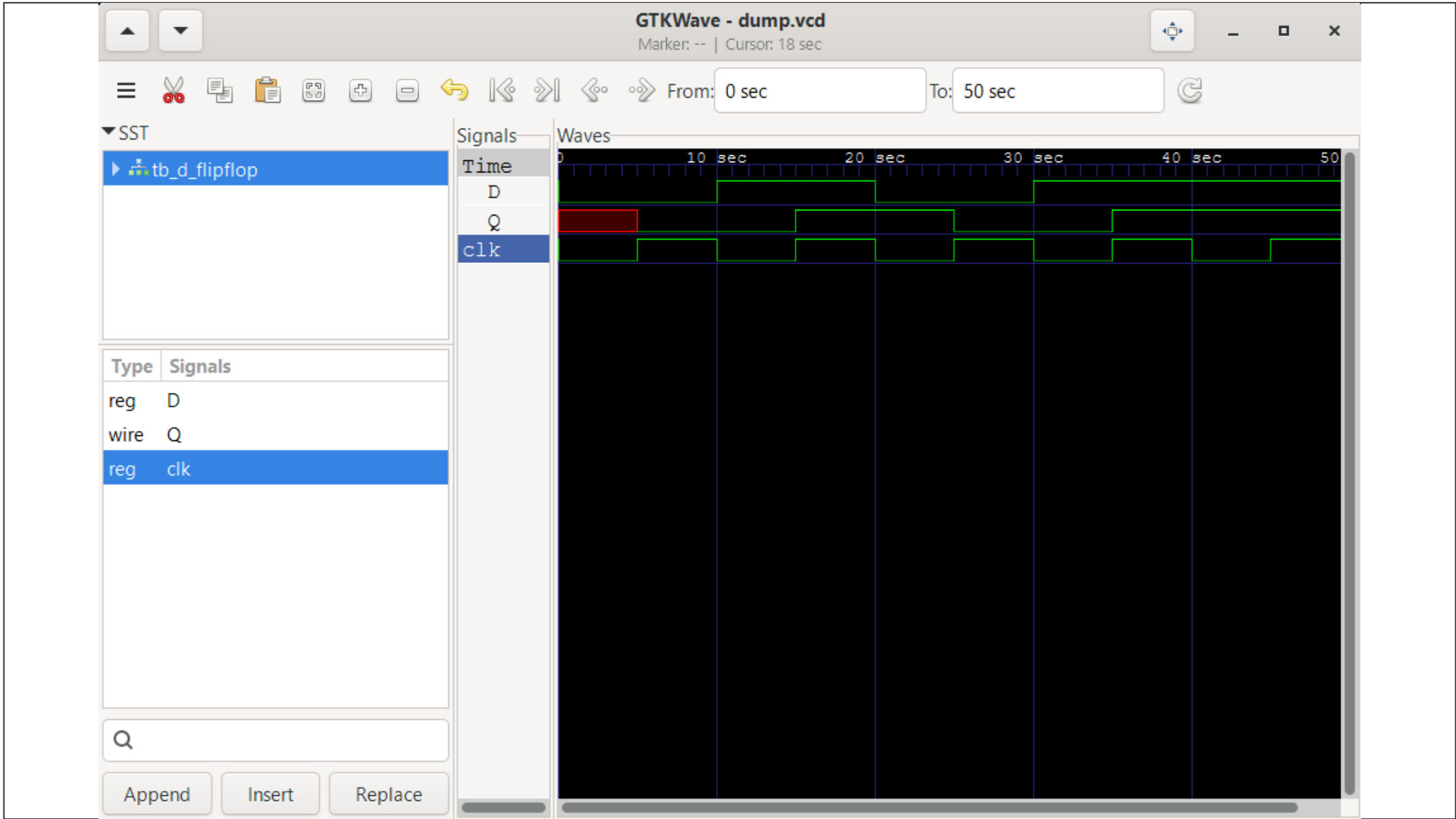


Fig 3: VCD waveform dump tracking Clock, D input, and Q output for D Flip-Flop.

Part B: JK Flip-Flop

1. RTL Block Diagram

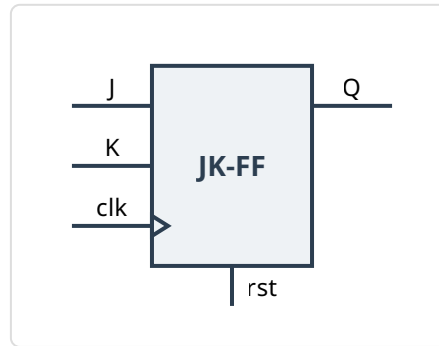


Fig 4: RTL Block Diagram representing the inputs and outputs of the JK Flip-Flop.

2. Verilog RTL Design & Simulation Console (VS Code)

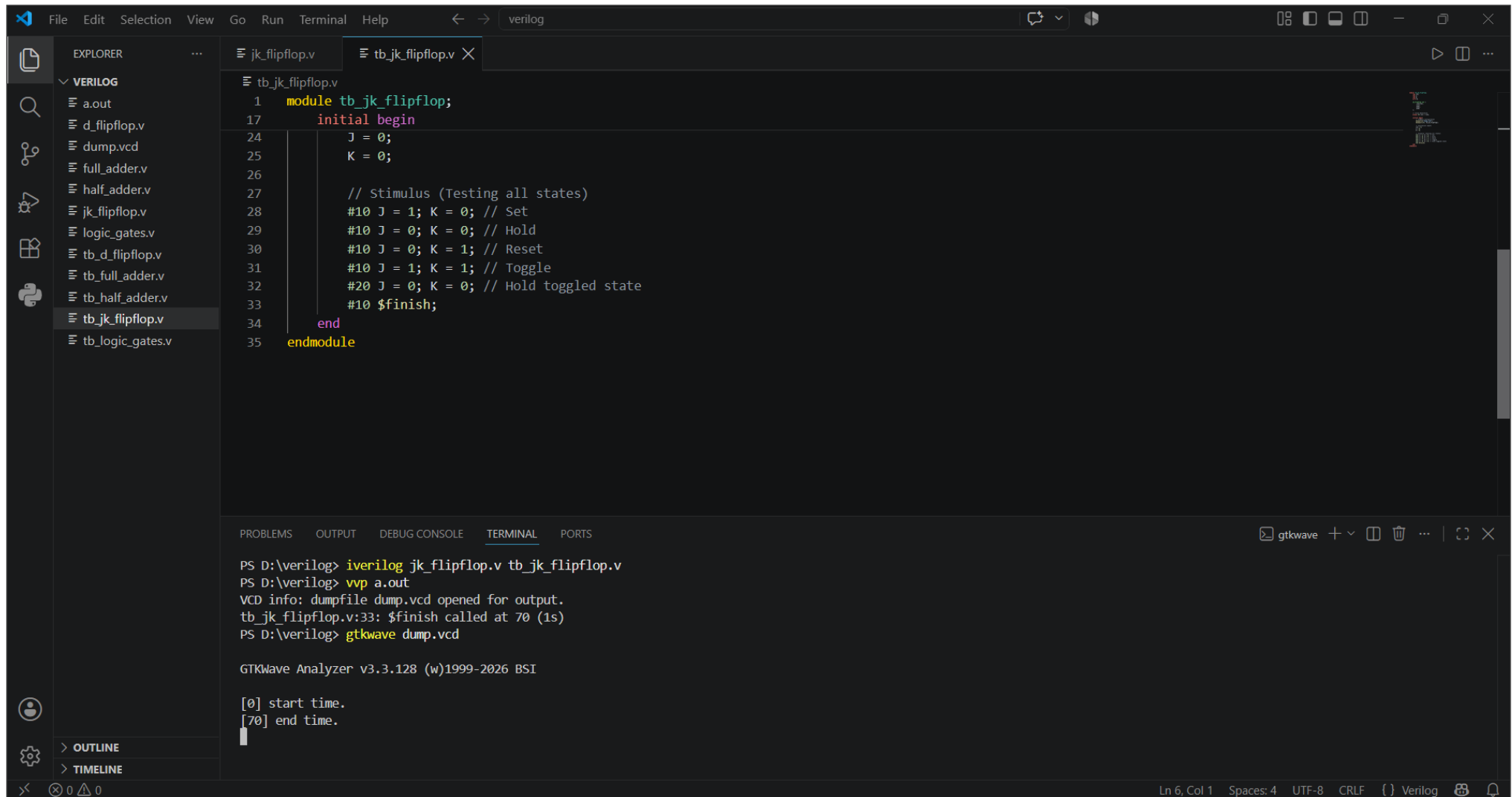


Fig 5: RTL logic handling Set, Reset, Hold, and Toggle states and Terminal output.

3. GTKWave Waveform Verification

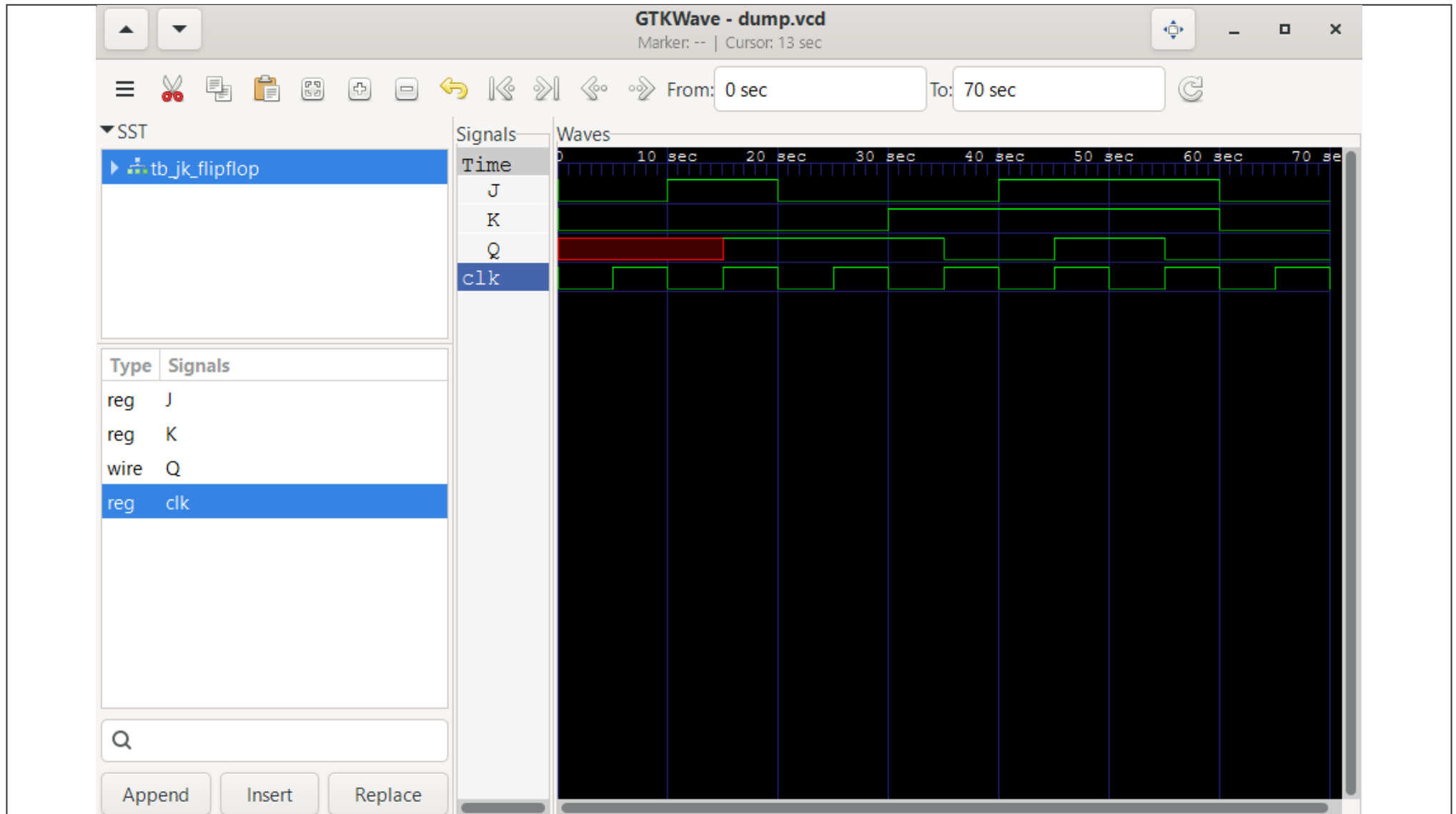


Fig 6: GTKWave verifying J and K inputs determining the Q output states.

Part C: 4-bit Register

1. RTL Block Diagram

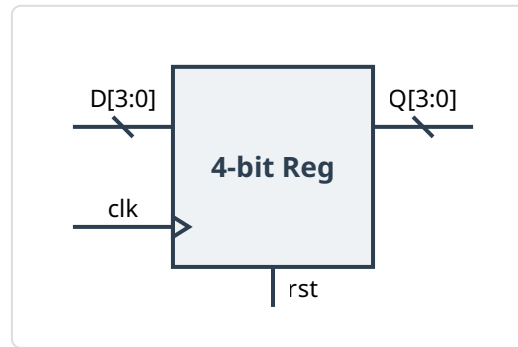


Fig 7: RTL Block Diagram representing the 4-bit parallel Register.

2. Verilog RTL Design & Simulation Console (VS Code)

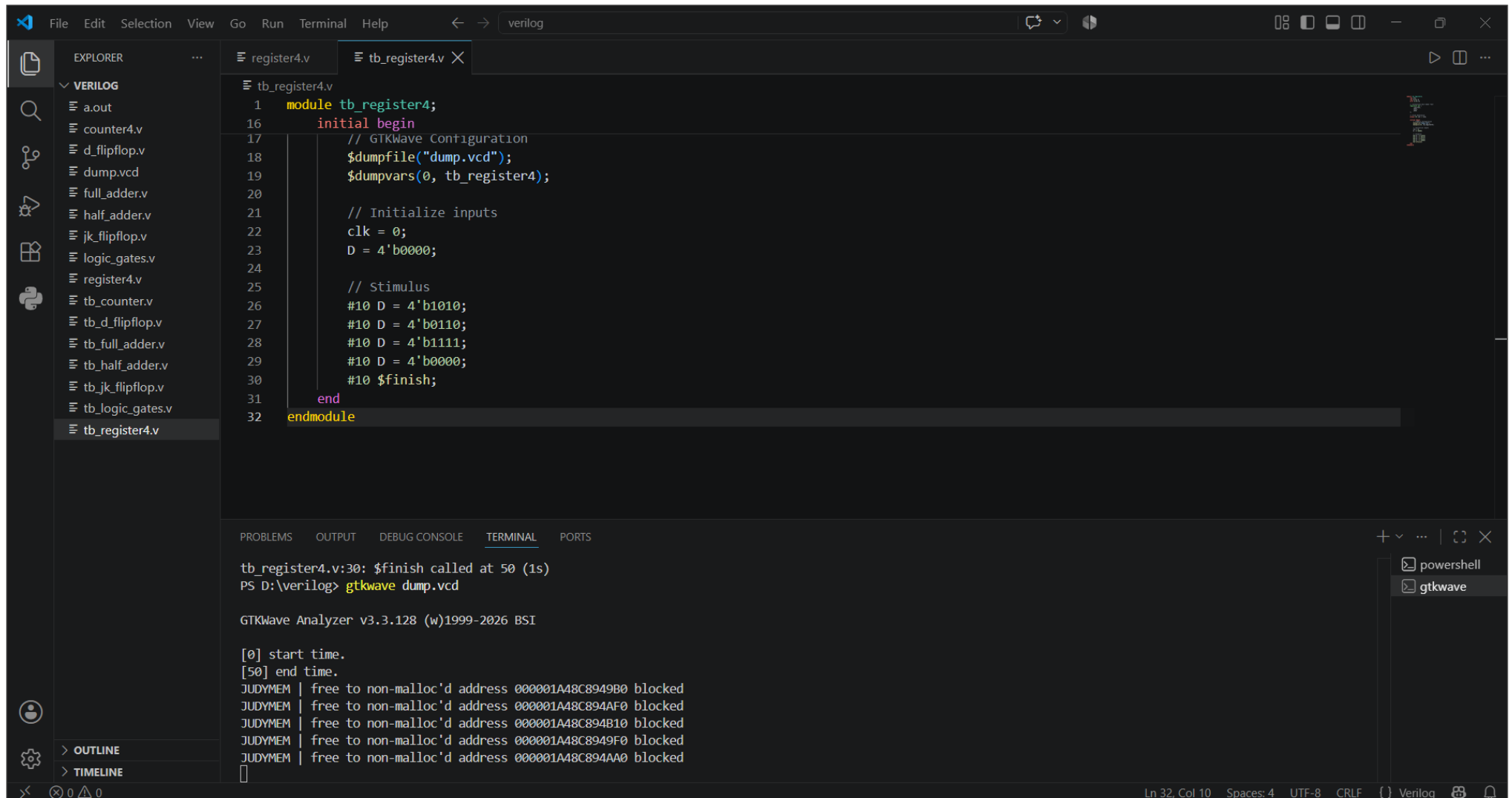


Fig 8: 4-bit Register RTL modeling and Terminal output verification.

3. GTKWave Waveform Verification

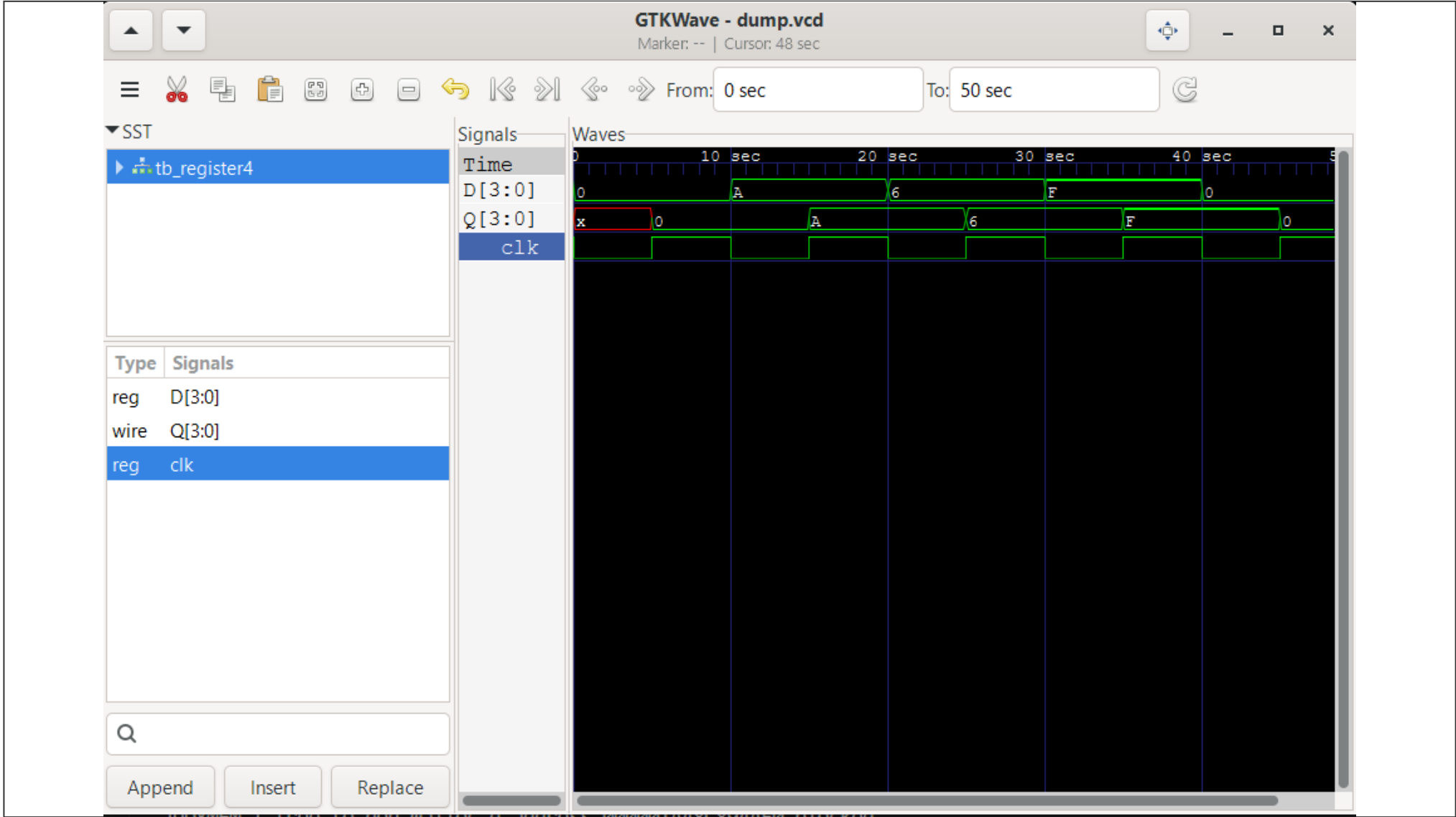


Fig 9: GTKWave tracking parallel data load and synchronous reset.

Part D: 4-bit Counter

1. RTL Block Diagram

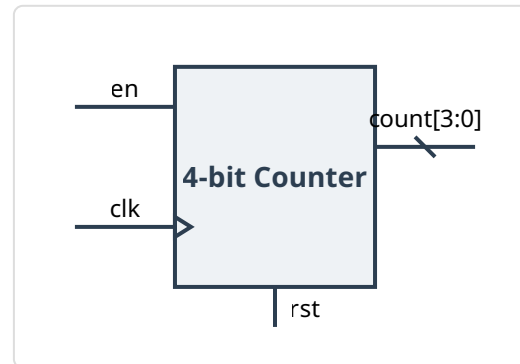


Fig 10: RTL Block Diagram representing the Parameterized 4-bit Counter.

2. Verilog RTL Design & Simulation Console (VS Code)

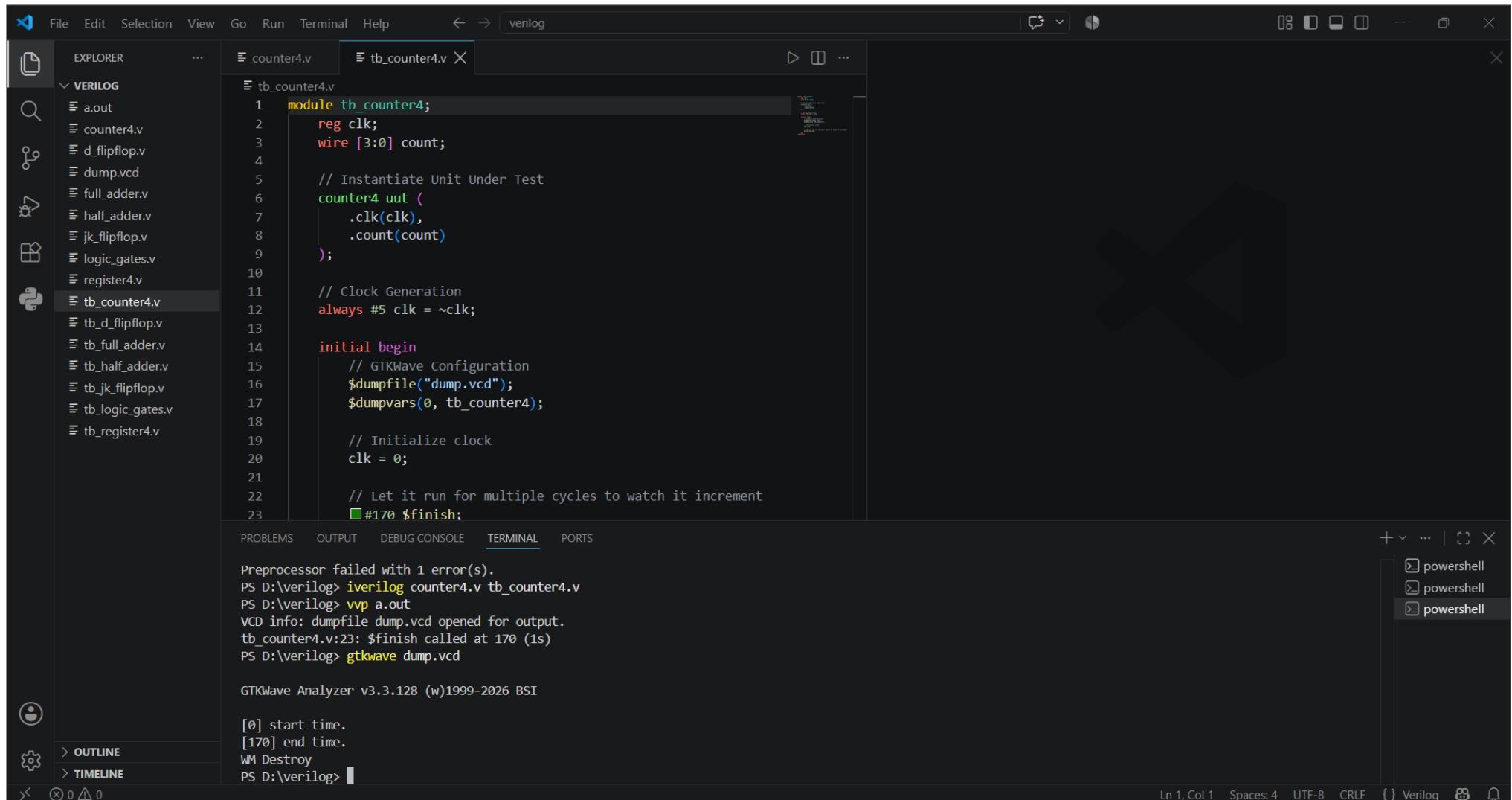


Fig 11: Parameterized 4-bit Counter RTL implementation and Terminal verification.

3. GTKWave Waveform Verification

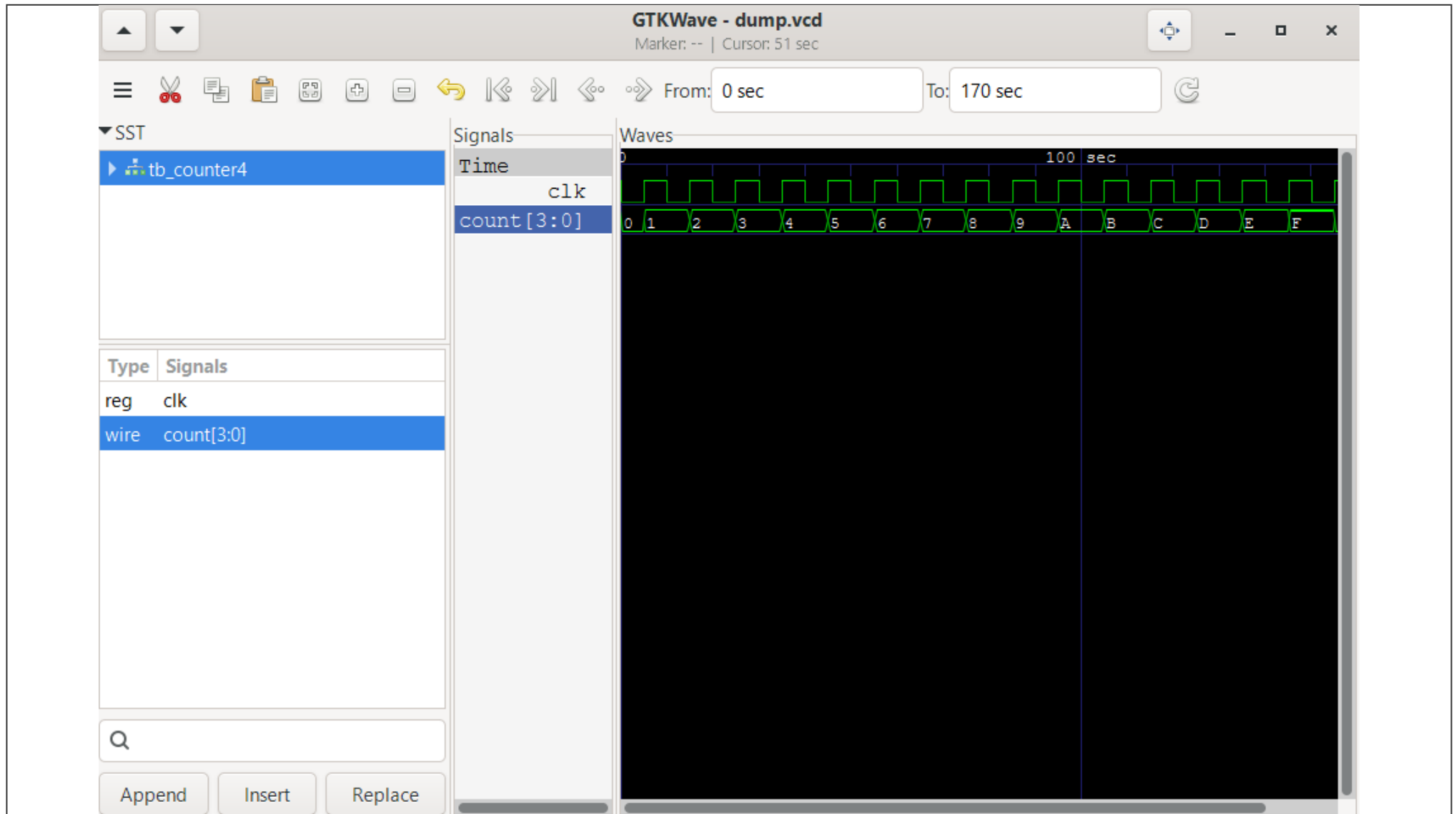


Fig 12: VCD output reflecting correct binary counting sequence and rollover.

4. KEY TAKEAWAYS & CONCLUSION

- **Functional Correctness:** All `always @(posedge clk)` blocks successfully modeled sequential logic, updating outputs accurately in synchronization with the clock edge.
- **Simulation Workflow:** Utilized `$dumpfile` and `$dumpvars` to generate VCD files for real-time visual inspection of clock cycles and state changes in GTKWave.
- **Task Completion:** RTL modules, testbenches, circuit diagrams, console logs, and visual waveform verifications successfully satisfy all mandatory and optional deliverables for Task 3.